HIGH LEVEL DESIGN DOCUMENT

MDE - Linux Onboarding (Manual)

OVERVIEW

1. Introduction

This document outlines the process of integrating Linux-based servers into the organization's centralized security monitoring platform. By deploying Microsoft Defender for Endpoint, we gain the ability to detect, investigate, and respond to advanced threats targeting our Linux infrastructure.

2. Purpose of the Document

The goal of this design is to provide a standardized approach for security administrators to manually install and configure the MDE agent. This ensures that even in environments without automated deployment tools, every Linux server maintains a consistent security posture.

3. Use of the Document

This document serves as:

- A step-by-step guide for basic manual implementation of MDE in Linux devices.
- A reference point for troubleshooting initial connectivity.
- A baseline for verifying that the agent is correctly communicating with the security portal.

4. In scope

- Manual installation of the MDE package on supported Linux Ubuntu distributions
- Configuration of the onboarding script provided by the Microsoft Defender portal.
- Verification of the "Active" status in the security dashboard.

5. Out of Scope

- Automated deployment via Ansible, Chef, or Puppet.
- Troubleshooting underlying hardware or OS-level performance issues unrelated to MDE.
- Configuration of advanced automated remediation or custom API integrations.

6. Overview

The manual onboarding process involves three primary phases:

1. **Preparation:** Ensuring the server has internet access to the necessary Microsoft endpoints.
2. **Installation:** Downloading the MDE software package and the unique organization onboarding script.
3. **Validation:** Running a connectivity test and a "mock" detection test to ensure the server is reporting data correctly.

7. Limitations of MDE on Linux devices

Features Not Supported on Linux. Below are few of the important MDE features which are not supported in Linux.

- **Attack Surface Reduction (ASR) rules**

These rules (e.g., blocking Office macros, preventing credential theft) are Windows-only.

- **Device Control (USB restrictions, peripheral management)**

No native enforcement on Linux.

- **Application Control (AppLocker, Windows Defender Application Control)**

Not available for Linux distributions.

- **Automated Investigation and Remediation (AIR)**

Linux does not support automatic remediation workflows; investigations are manual.

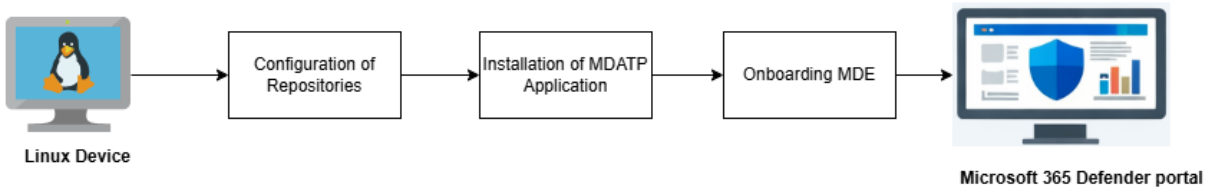
Exploit Protection (Windows Defender Exploit Guard)

- Not supported on Linux.

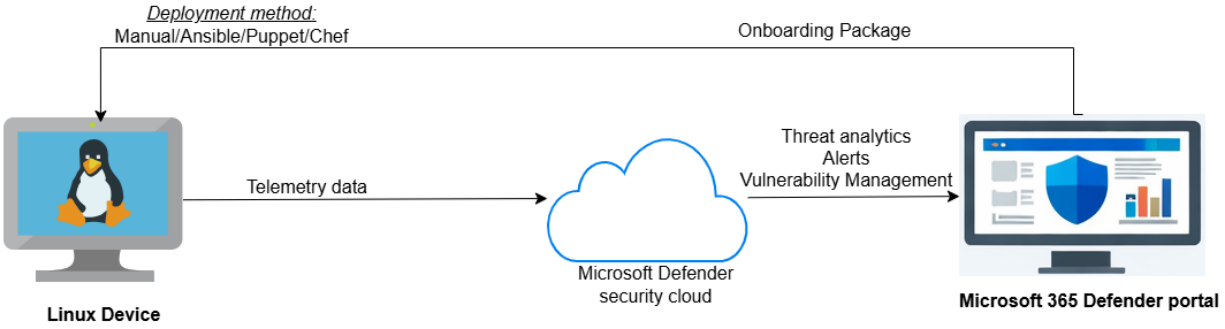
8. Architecture Diagram

The architecture follows a client-cloud model. The Linux server hosts the MDE agent, which communicates outbound to Microsoft's cloud security services.

Steps for Onboarding

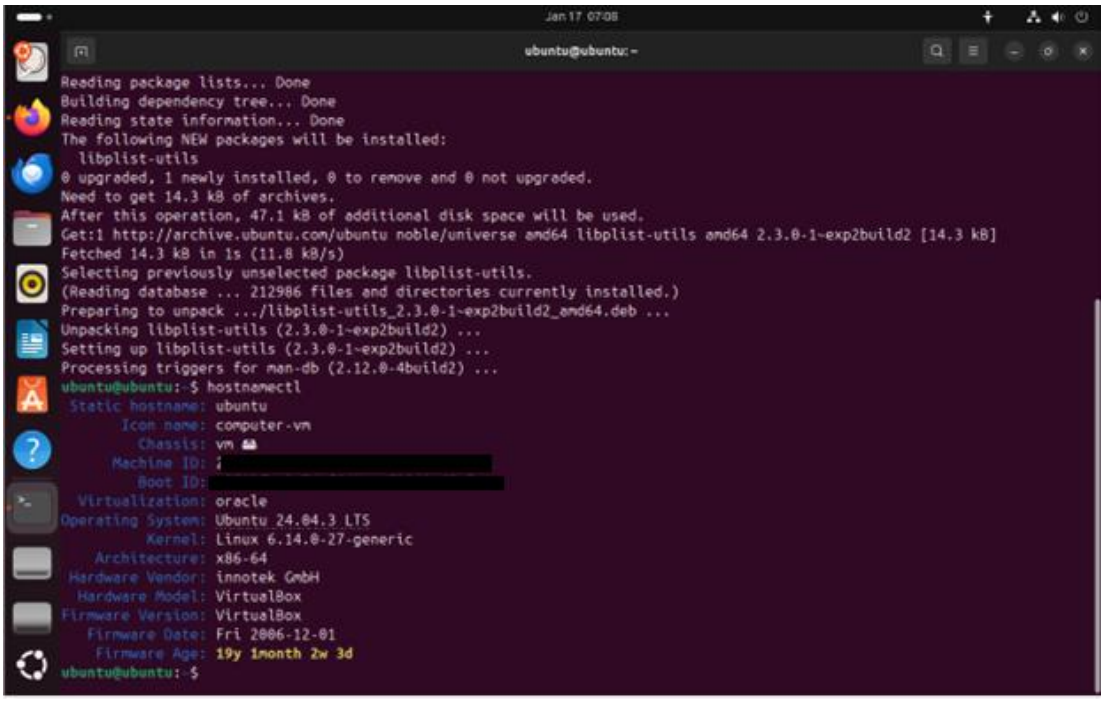


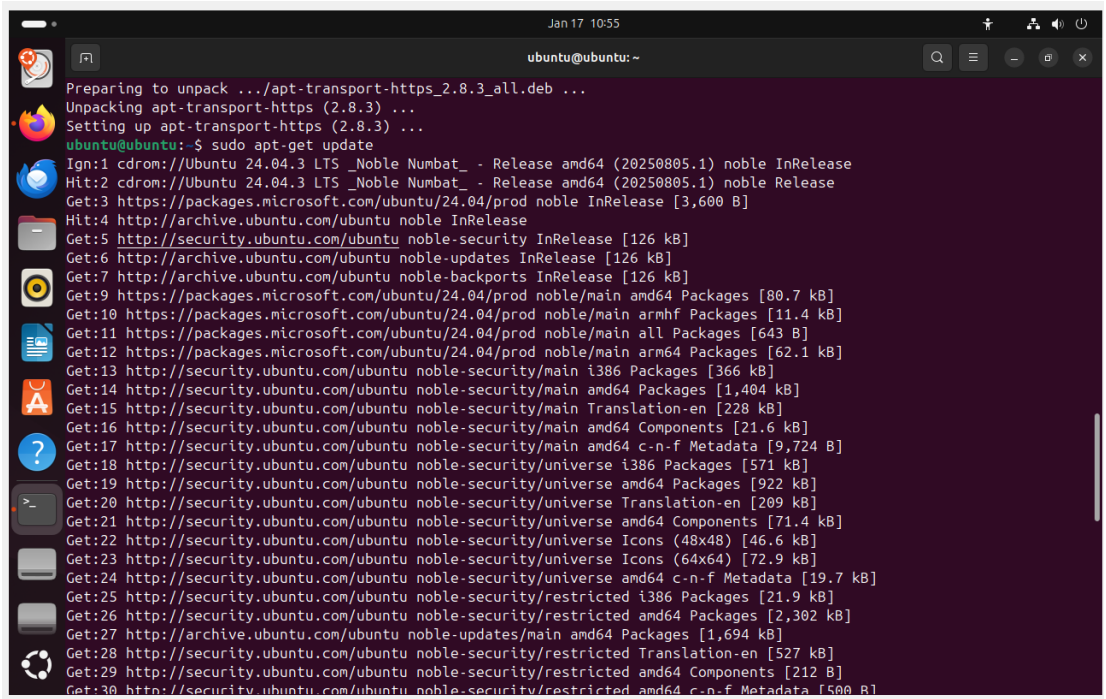
MDE Architecture



9. Screenshots of Implementation

Check Hostname and check if the device is up to date





Install curl

```
ubuntu@ubuntu: ~
ubuntu@ubuntu:~$ sudo apt install curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  curl
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 226 kB of archives.
After this operation, 534 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 curl amd64 8.5.0
-2ubuntu10.6 [226 kB]
Fetched 226 kB in 2s (99.4 kB/s)
Selecting previously unselected package curl.
(Reading database ... 212979 files and directories currently installed.)
Preparing to unpack .../curl_8.5.0-2ubuntu10.6_amd64.deb ...
Unpacking curl (8.5.0-2ubuntu10.6) ...
Setting up curl (8.5.0-2ubuntu10.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
ubuntu@ubuntu:~$
```

Note: The device does not have restriction on internet connectivity.

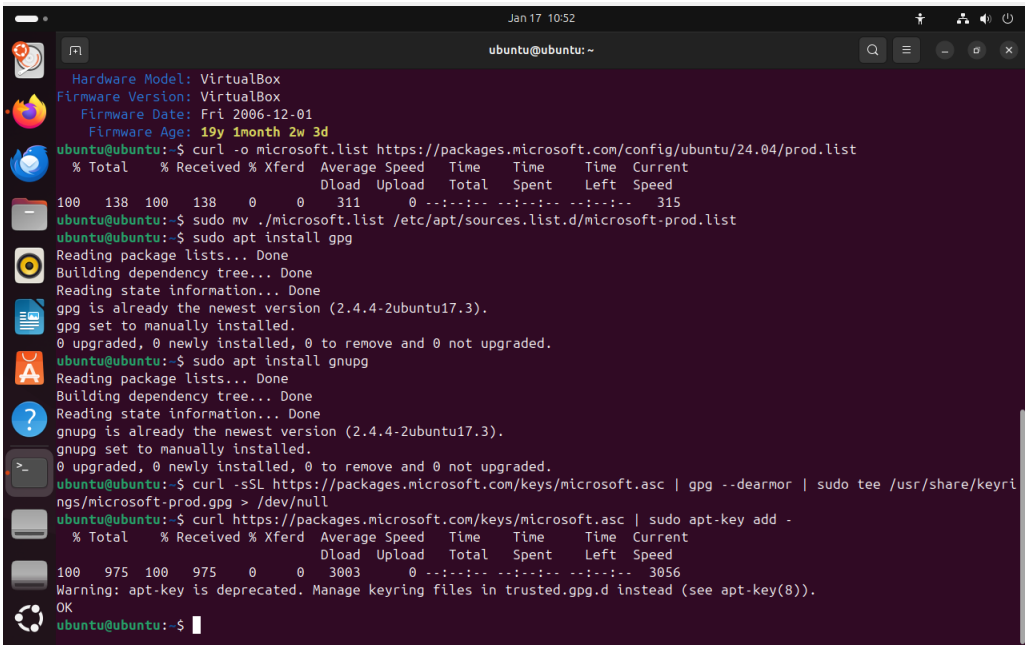
Install libplist-utils

```
ubuntu@ubuntu: ~
Unpacking curl (8.5.0-2ubuntu10.6) ...
Setting up curl (8.5.0-2ubuntu10.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
ubuntu@ubuntu:~$ sudo apt install libplist-utils
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  libplist-utils
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 14.3 kB of archives.
After this operation, 47.1 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble/universe amd64 libplist-utils amd64
 2.3.0-1~exp2build2 [14.3 kB]
Fetched 14.3 kB in 1s (11.8 kB/s)
Selecting previously unselected package libplist-utils.
(Reading database ... 212986 files and directories currently installed.)
Preparing to unpack .../libplist-utils_2.3.0-1~exp2build2_amd64.deb ...
Unpacking libplist-utils (2.3.0-1~exp2build2) ...
Setting up libplist-utils (2.3.0-1~exp2build2) ...
Processing triggers for man-db (2.12.0-4build2) ...
ubuntu@ubuntu:~$ S
```

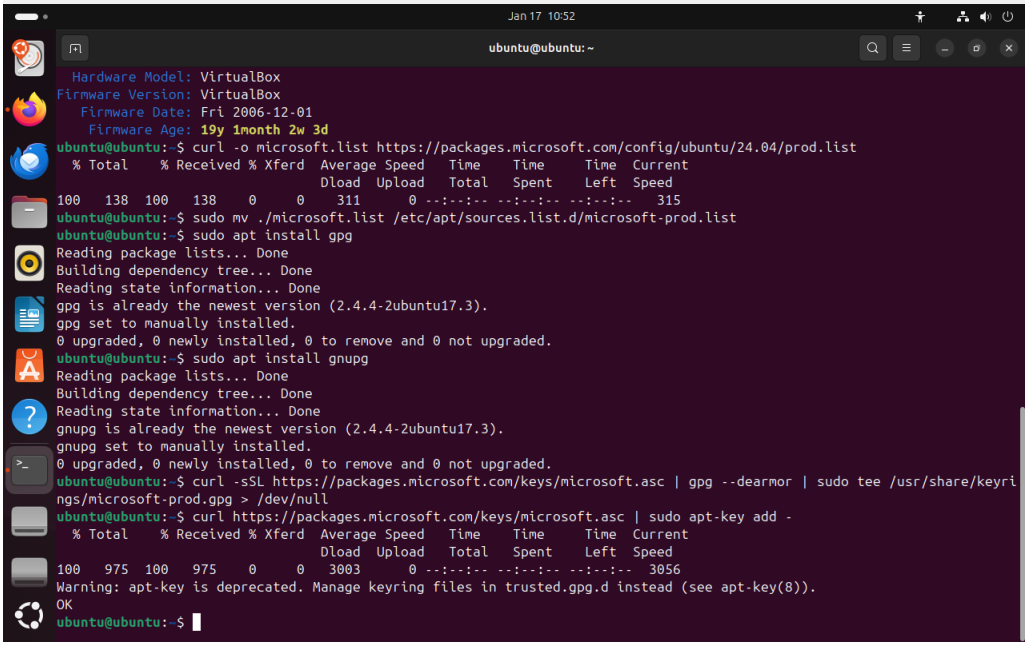
In the following command, replace *[distro]* and *[version]* with the information you've identified:

curl -o microsoft.list
https://packages.microsoft.com/config/[distro]/[version]/[channel].list

Example: curl -o microsoft.list
https://packages.microsoft.com/config/ubuntu/18.04/prod.list



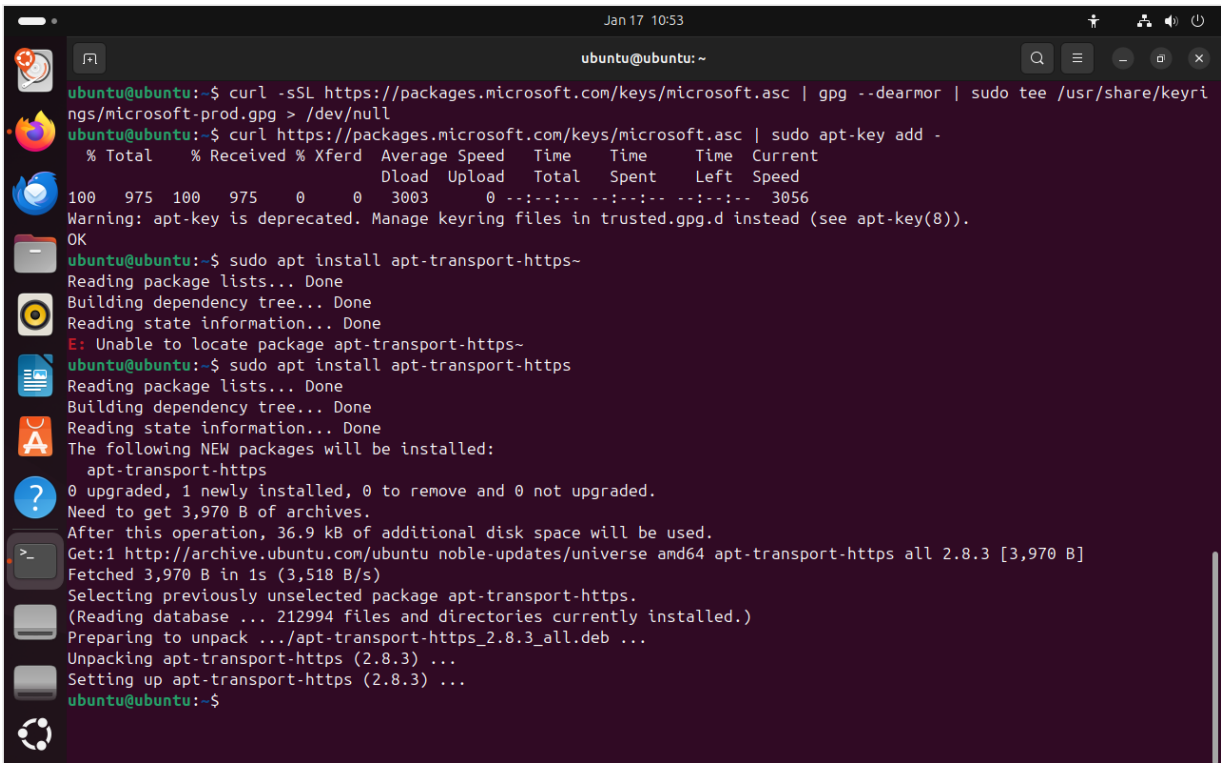
Install the gpg package if not already installed:
sudo apt install gpg



Install the Microsoft GPG public key:

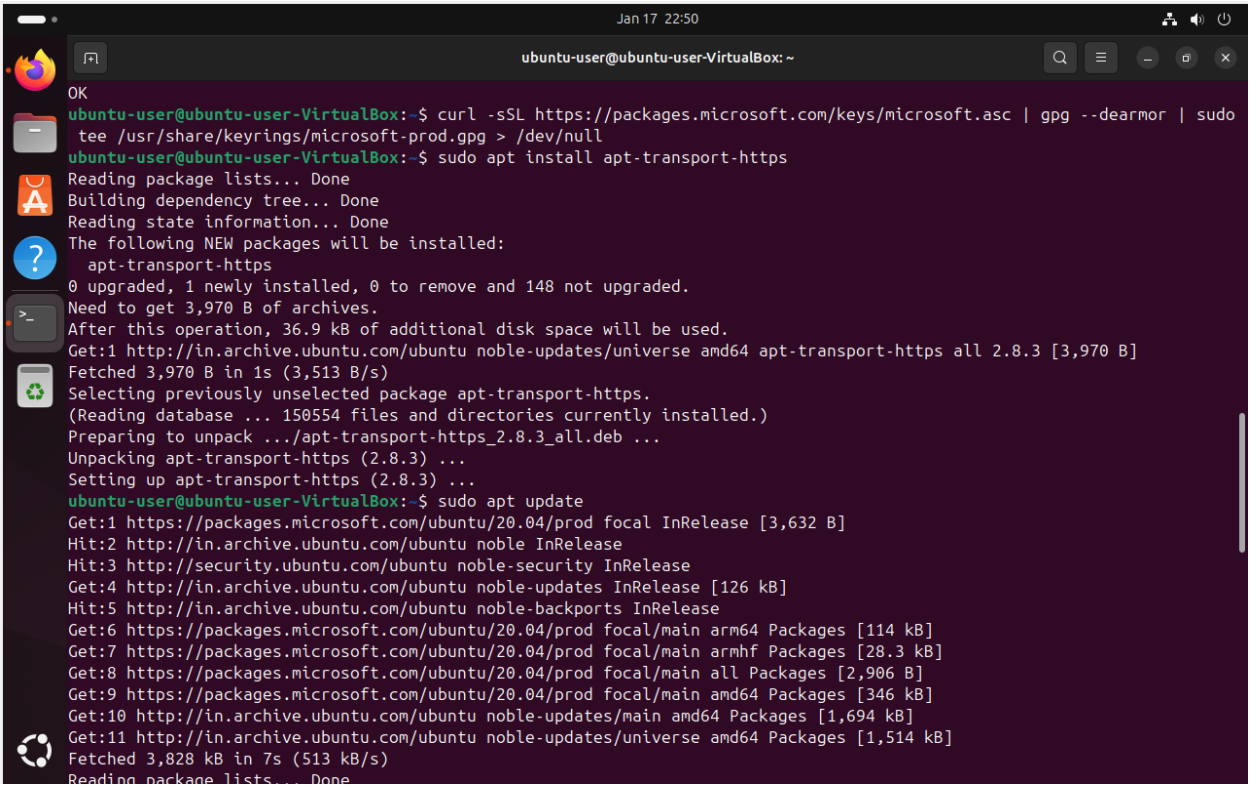
curl -sSL https://packages.microsoft.com/keys/microsoft.asc | gpg --dearmor | sudo tee /etc/apt/trusted.gpg.d/microsoft.gpg > /dev/null

sudo chmod o+r /etc/apt/trusted.gpg.d/microsoft.gpg



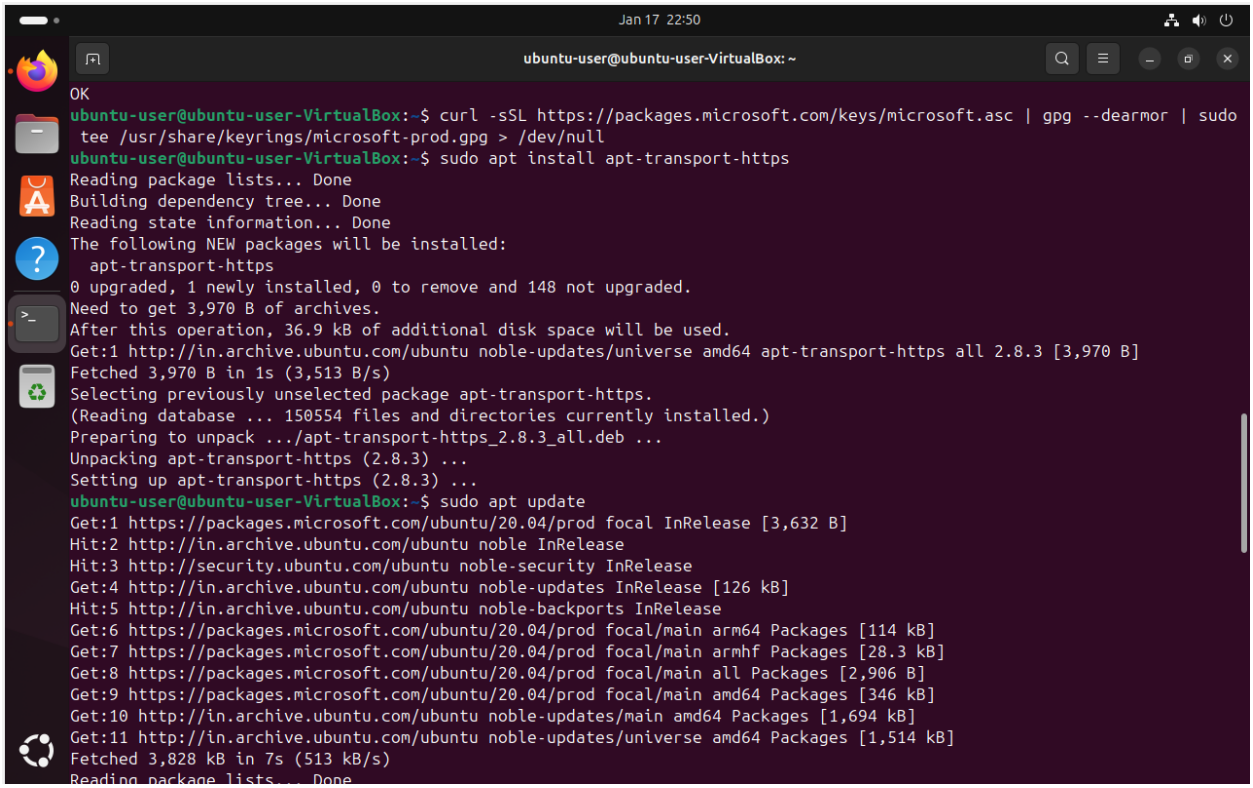
Install the HTTPS driver if not already installed:

sudo apt install apt-transport-https



9.Update the repository metadata:

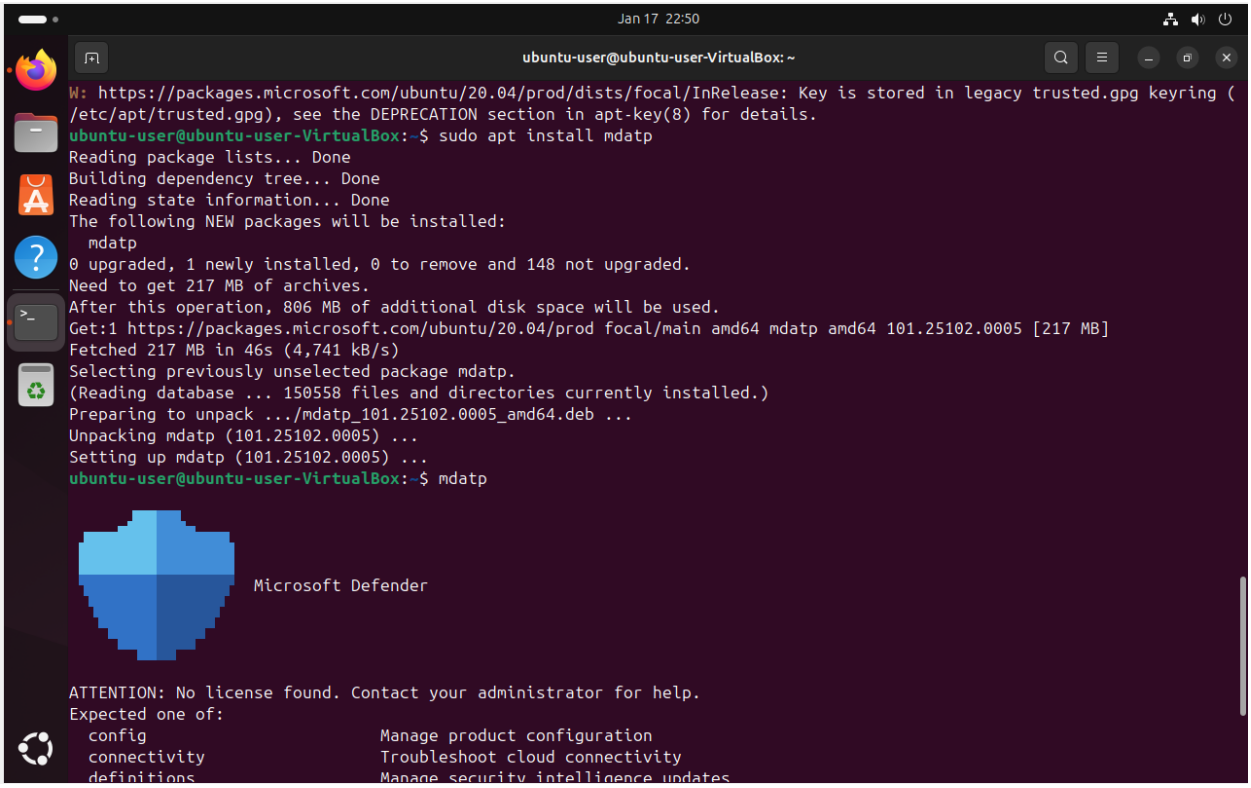
sudo apt update

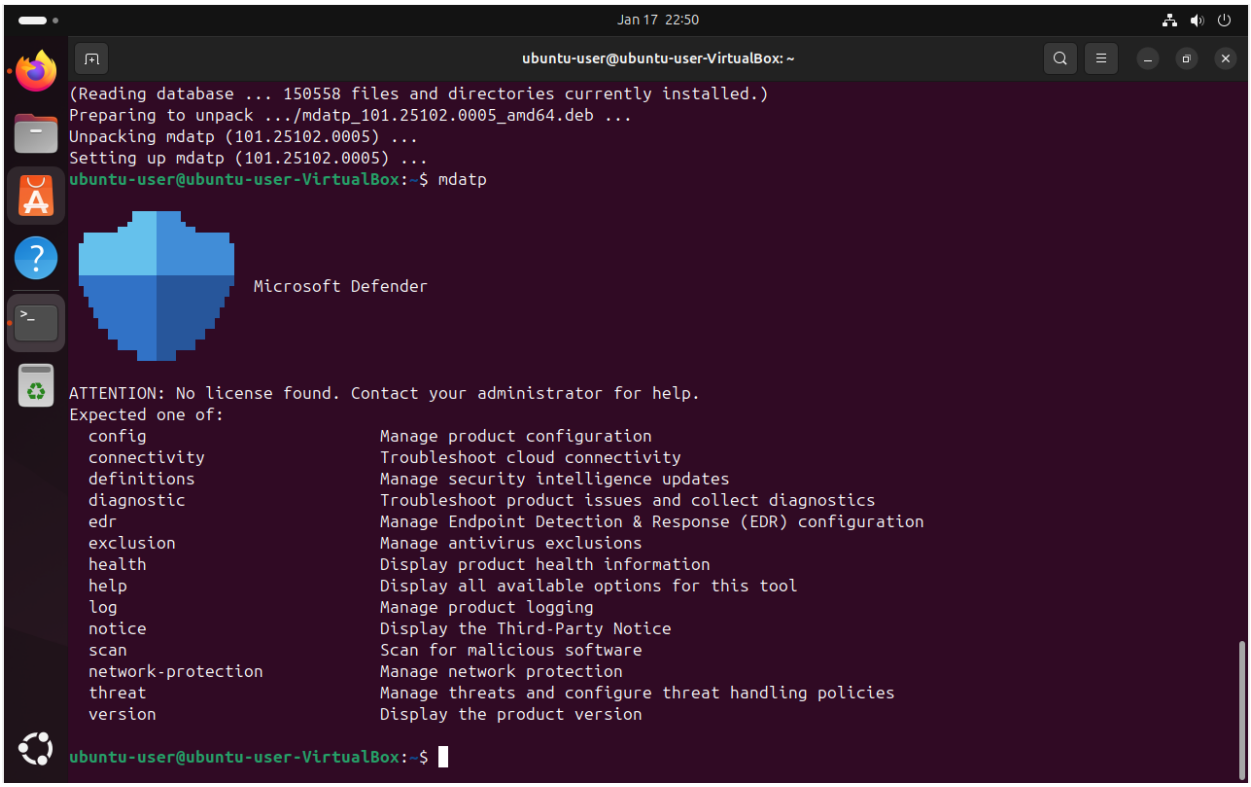


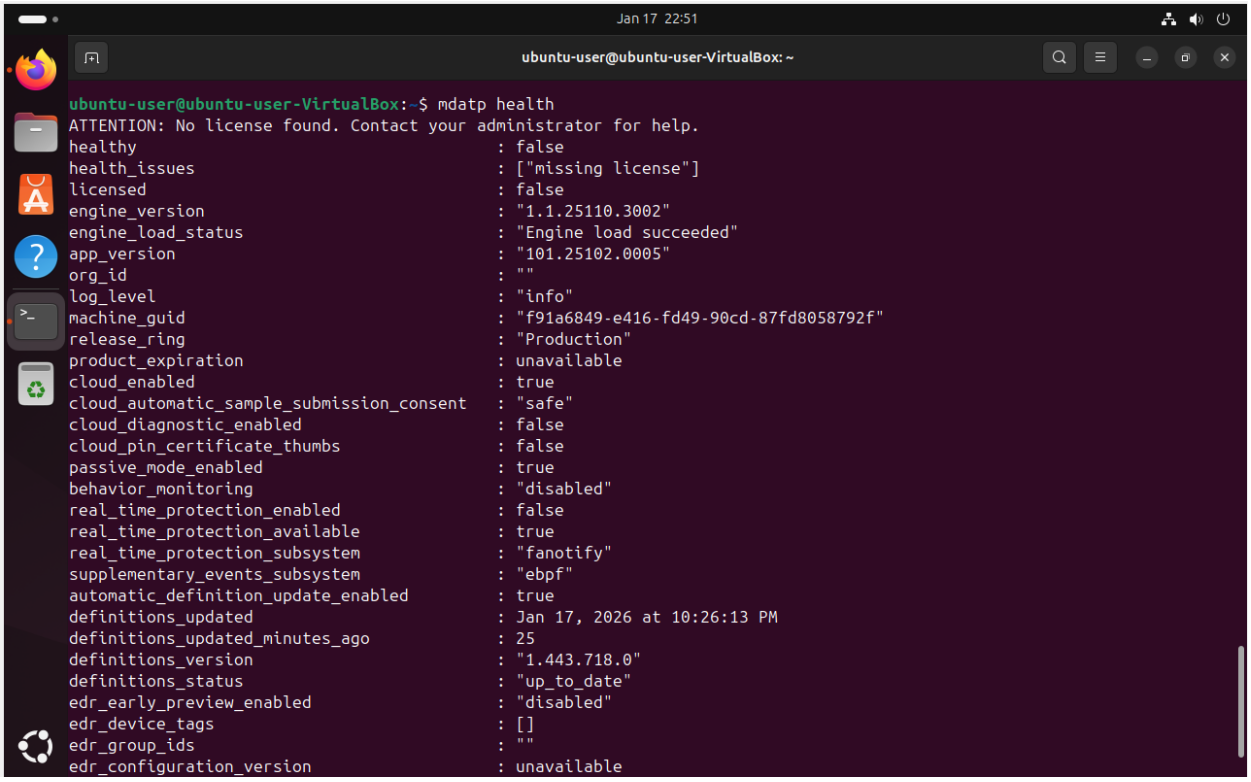
Application installation

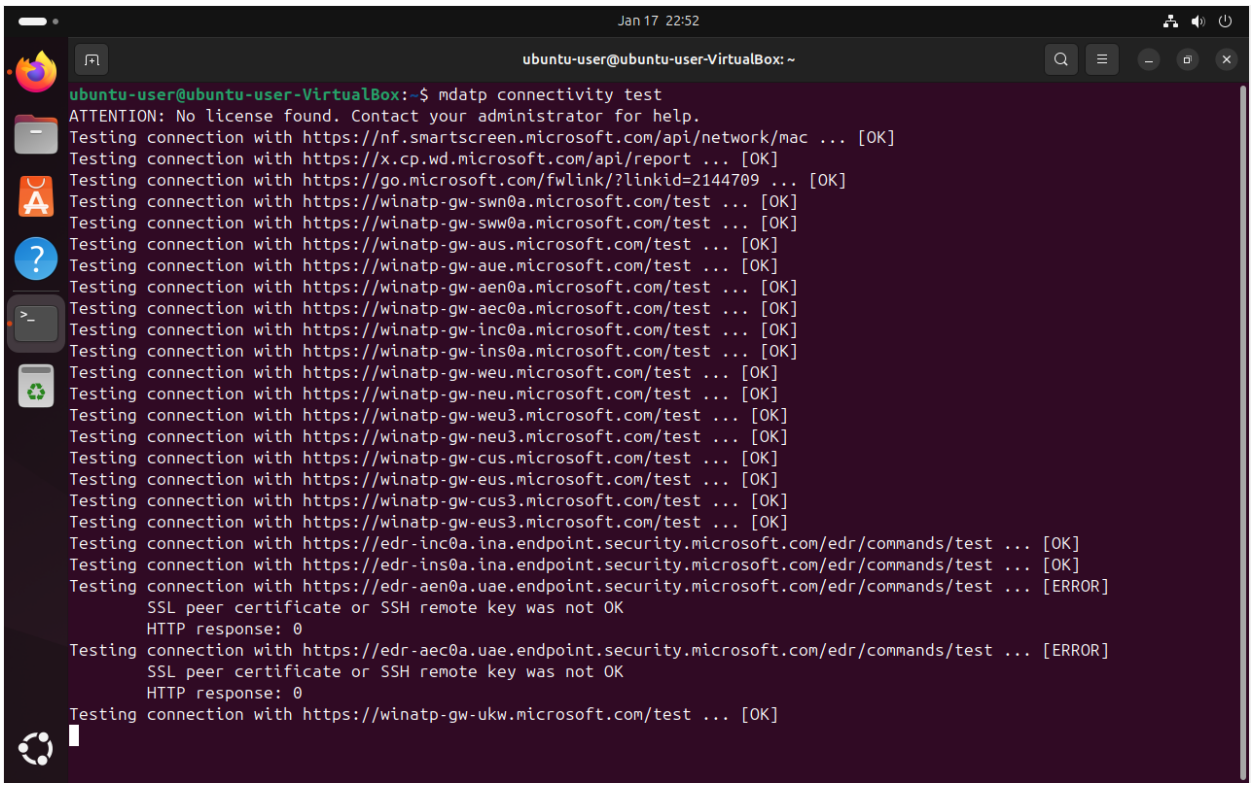
Use the commands in the following sections to install Defender for Endpoint on your Linux distribution.

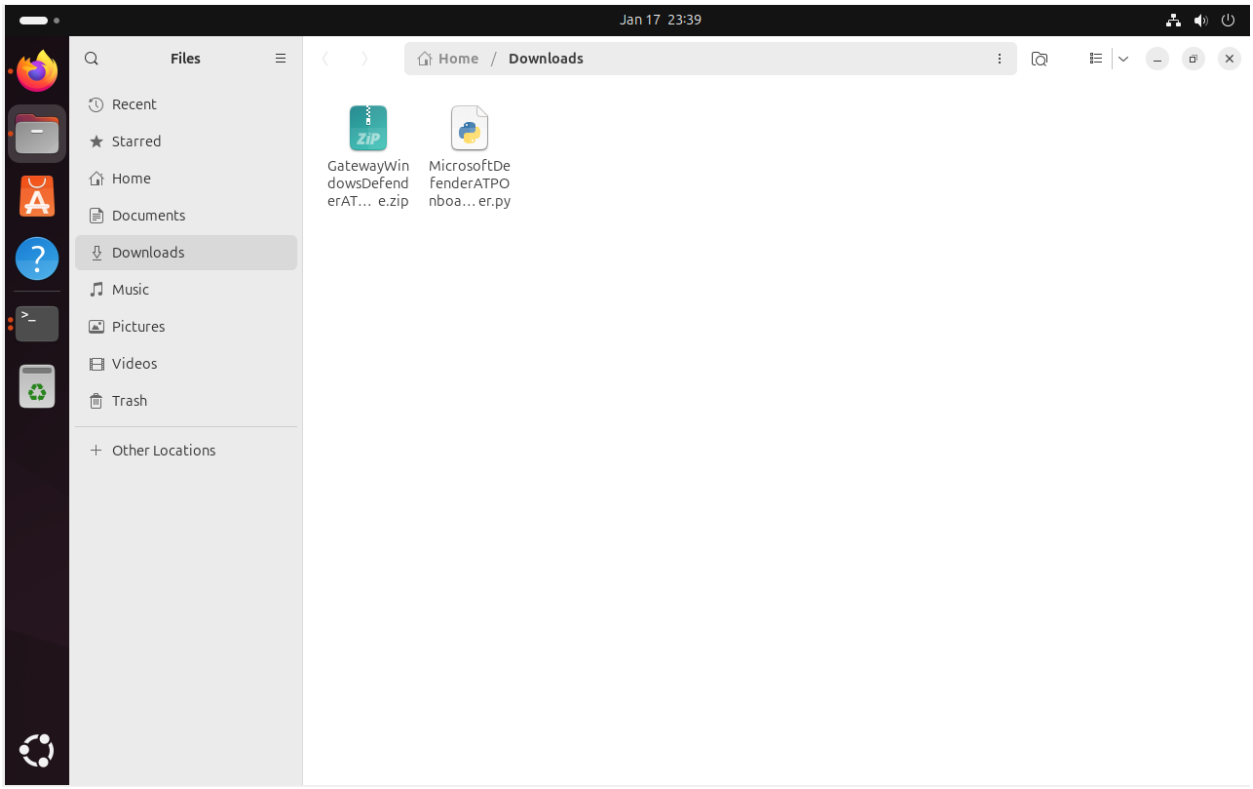
sudo apt install mdatp

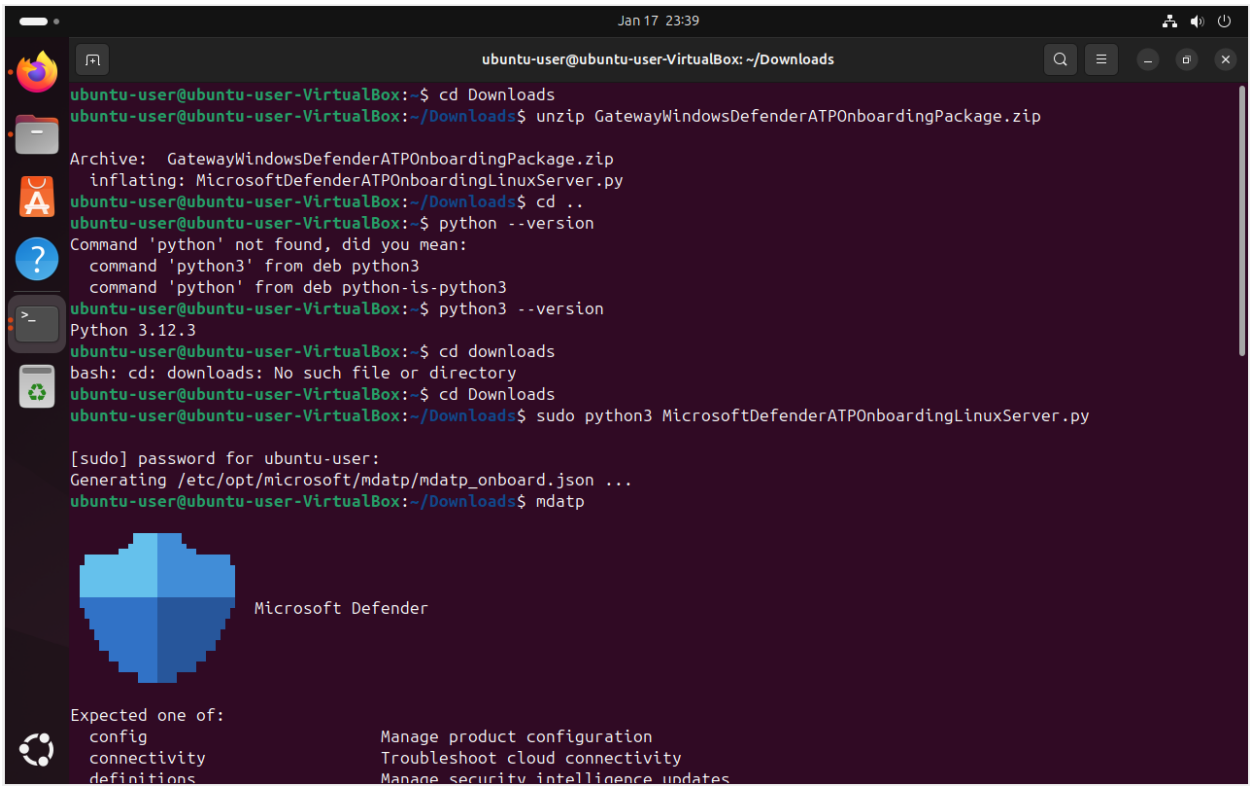

Jan 17 22:50
ubuntu-user@ubuntu-user-VirtualBox: ~
W: https://packages.microsoft.com/ubuntu/20.04/prod/dists/focal/InRelease: Key is stored in legacy trusted.gpg keyring (/etc/apt/trusted.gpg), see the DEPRECATION section in apt-key(8) for details.
ubuntu-user@ubuntu-user-VirtualBox:~$ sudo apt install mdatp
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
mdatp
0 upgraded, 1 newly installed, 0 to remove and 148 not upgraded.
Need to get 217 MB of archives.
After this operation, 806 MB of additional disk space will be used.
Get:1 https://packages.microsoft.com/ubuntu/20.04/prod focal/main amd64 mdatp amd64 101.25102.0005 [217 MB]
Fetched 217 MB in 46s (4,741 kB/s)
Selecting previously unselected package mdatp.
(Reading database ... 150558 files and directories currently installed.)
Preparing to unpack .../mdatp_101.25102.0005_amd64.deb ...
Unpacking mdatp (101.25102.0005) ...
Setting up mdatp (101.25102.0005) ...
ubuntu-user@ubuntu-user-VirtualBox:~$ mdatp
Microsoft Defender
ATTENTION: No license found. Contact your administrator for help.
Expected one of:
config Manage product configuration
connectivity Troubleshoot cloud connectivity


Jan 17 22:50
ubuntu-user@ubuntu-user-VirtualBox: ~
(Reading database ... 150558 files and directories currently installed.)
Preparing to unpack .../mdatp_101.25102.0005_amd64.deb ...
Unpacking mdatp (101.25102.0005) ...
Setting up mdatp (101.25102.0005) ...
ubuntu-user@ubuntu-user-VirtualBox:~$ mdatp
Microsoft Defender
ATTENTION: No license found. Contact your administrator for help.
Expected one of:
config Manage product configuration
connectivity Troubleshoot cloud connectivity
definitions Manage security intelligence updates
diagnostic Troubleshoot product issues and collect diagnostics
edr Manage Endpoint Detection & Response (EDR) configuration
exclusion Manage antivirus exclusions
health Display product health information
help Display all available options for this tool
log Manage product logging
notice Display the Third-Party Notice
scan Scan for malicious software
network-protection Manage network protection
threat Manage threats and configure threat handling policies
version Display the product version
ubuntu-user@ubuntu-user-VirtualBox:~$

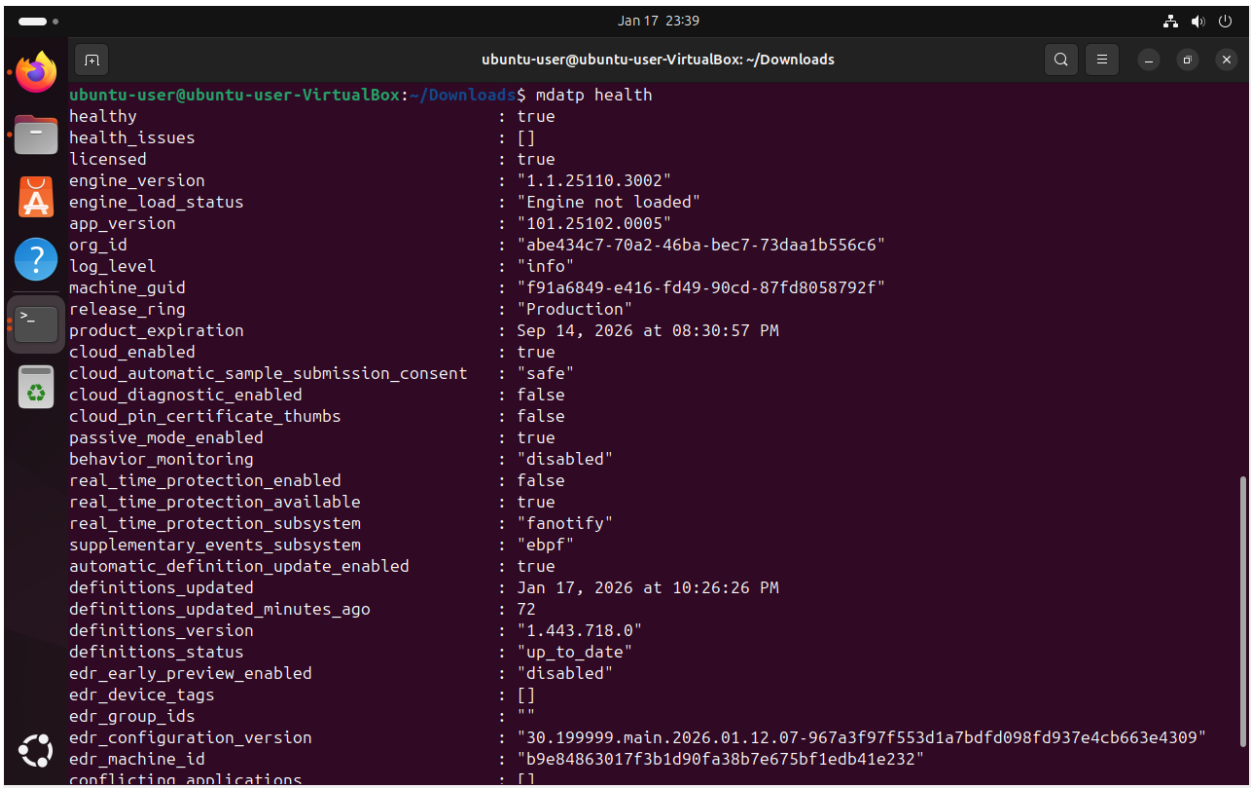

Jan 17 22:51
ubuntu-user@ubuntu-user-VirtualBox: ~
ubuntu-user@ubuntu-user-VirtualBox:~$ mdatp health
ATTENTION: No license found. Contact your administrator for help.
healthy : false
health_issues : ["missing license"]
licensed : false
engine_version : "1.1.25110.3002"
engine_load_status : "Engine load succeeded"
app_version : "101.25102.0005"
org_id : ""
log_level : "info"
machine_guid : "f91a6849-e416-fd49-90cd-87fd8058792f"
release_ring : "Production"
product_expiration : unavailable
cloud_enabled : true
cloud_automatic_sample_submission_consent : "safe"
cloud_diagnostic_enabled : false
cloud_pin_certificate_thumbs : false
passive_mode_enabled : true
behavior_monitoring : "disabled"
real_time_protection_enabled : false
real_time_protection_available : true
real_time_protection_subsystem : "fanotify"
supplementary_events_subsystem : "ebpf"
automatic_definition_update_enabled : true
definitions_updated : Jan 17, 2026 at 10:26:13 PM
definitions_updated_minutes_ago : 25
definitions_version : "1.443.718.0"
definitions_status : "up_to_date"
edr_early_preview_enabled : "disabled"
edr_device_tags : []
edr_group_ids : ""
edr_configuration_version : unavailable


Jan 17 22:52
ubuntu-user@ubuntu-user-VirtualBox: ~
ubuntu-user@ubuntu-user-VirtualBox:~$ mdatp connectivity test
ATTENTION: No license found. Contact your administrator for help.
Testing connection with https://nf.smartscreen.microsoft.com/api/network/mac ... [OK]
Testing connection with https://x.cp.wd.microsoft.com/api/report ... [OK]
Testing connection with https://go.microsoft.com/fwlink/?linkid=2144709 ... [OK]
Testing connection with https://winatp-gw-swn0a.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-sww0a.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-aus.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-aue.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-aen0a.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-aec0a.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-inc0a.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-ins0a.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-weu.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-neu.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-weu3.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-neu3.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-cus.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-eus.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-cus3.microsoft.com/test ... [OK]
Testing connection with https://winatp-gw-eus3.microsoft.com/test ... [OK]
Testing connection with https://edr-inc0a.ina.endpoint.security.microsoft.com/edr/commands/test ... [OK]
Testing connection with https://edr-ins0a.ina.endpoint.security.microsoft.com/edr/commands/test ... [OK]
Testing connection with https://edr-aen0a.uae.endpoint.security.microsoft.com/edr/commands/test ... [ERROR]
SSL peer certificate or SSH remote key was not OK
HTTP response: 0
Testing connection with https://edr-aec0a.uae.endpoint.security.microsoft.com/edr/commands/test ... [ERROR]
SSL peer certificate or SSH remote key was not OK
HTTP response: 0
Testing connection with https://winatp-gw-ukw.microsoft.com/test ... [OK]

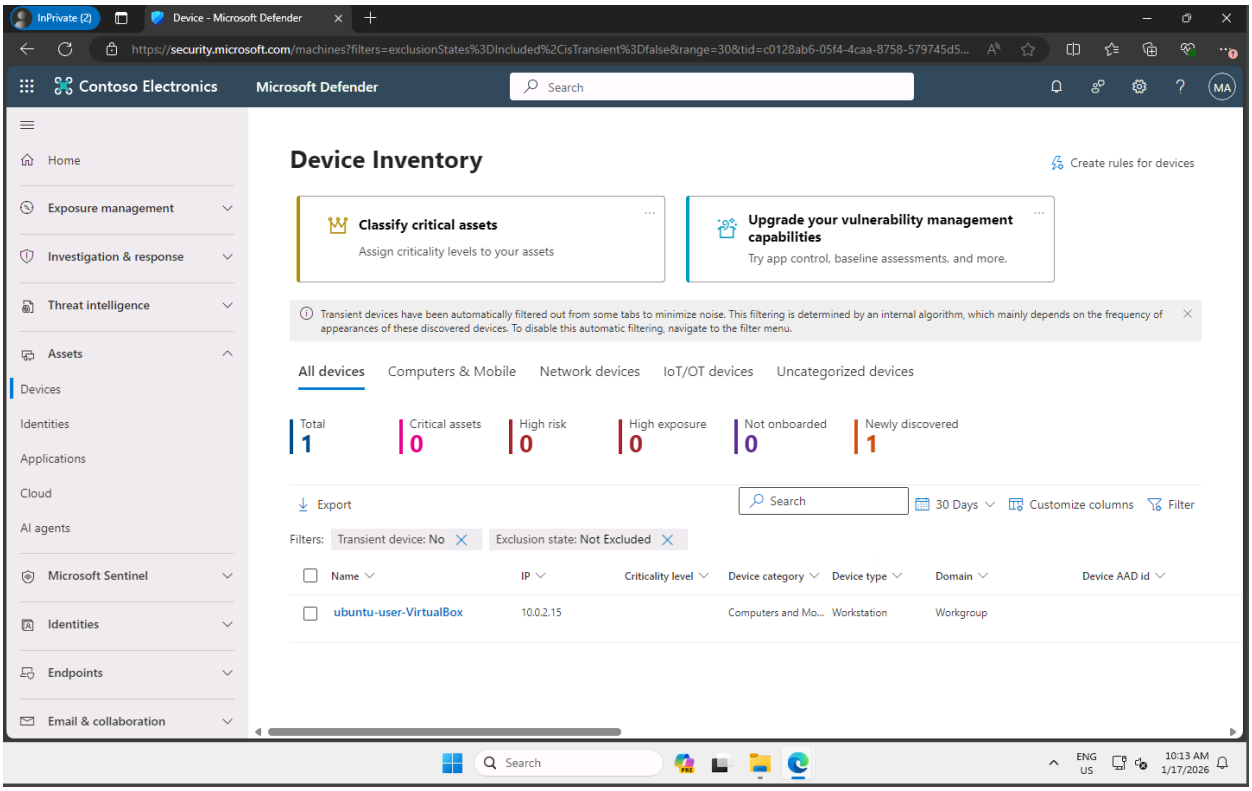

Jan 17 23:39
Files
Home / Downloads
Recent
Starred
Home
Documents
Downloads
Music
Pictures
Videos
Trash
Other Locations
GatewayWindowsDefenderAT... e.zip
MicrosoftDefenderATPOnboa... er.py

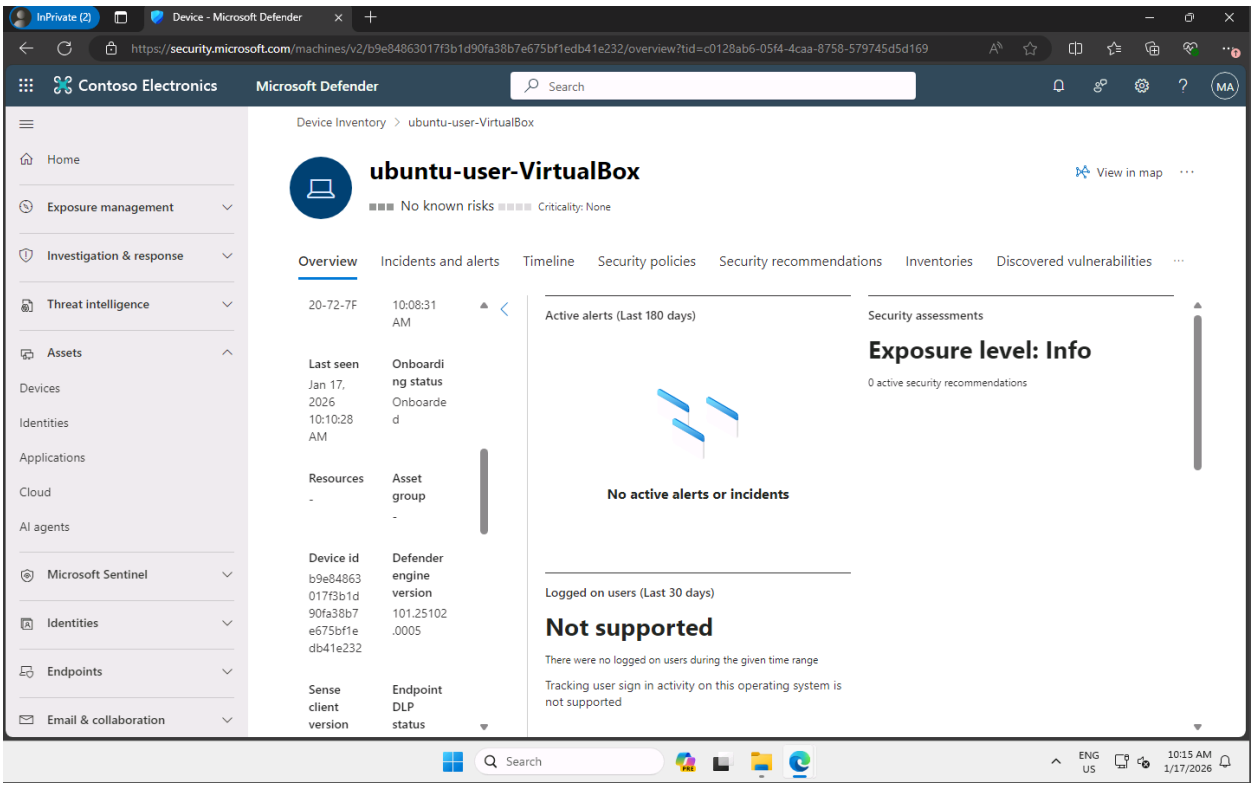


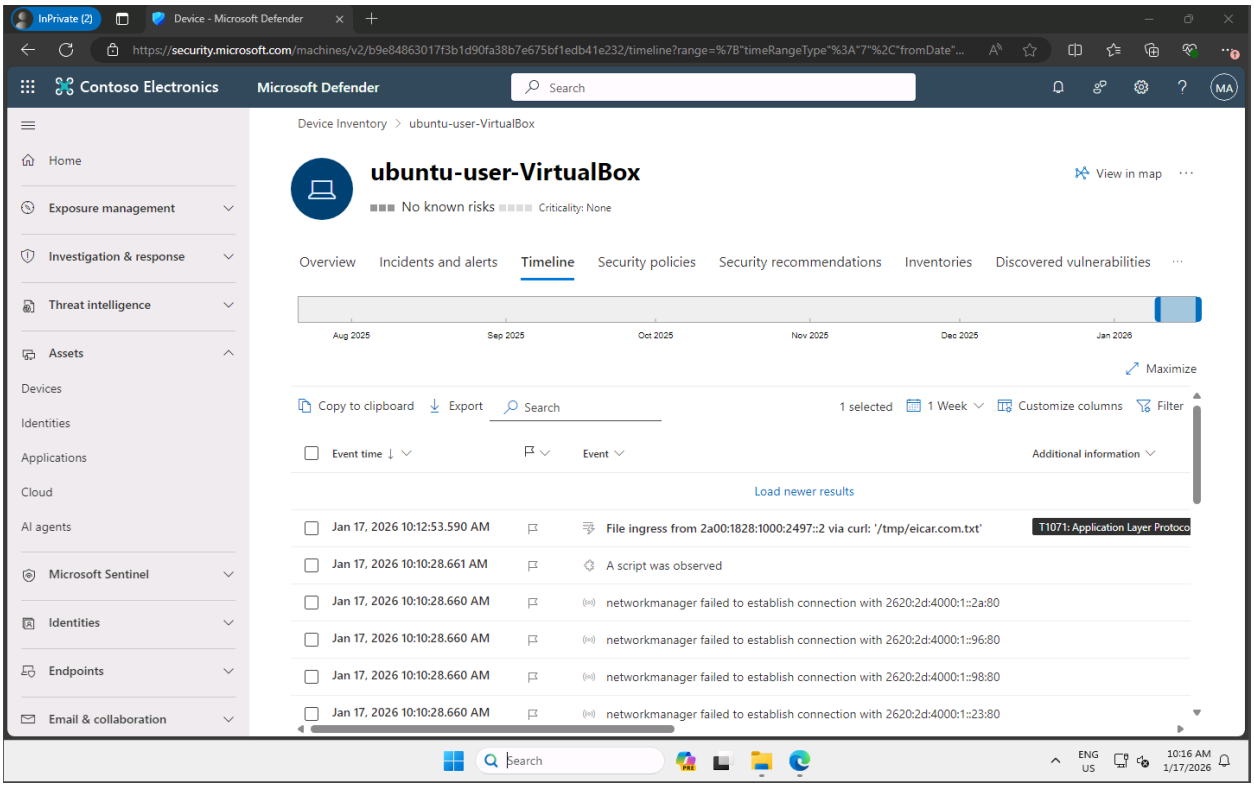
mdatp health

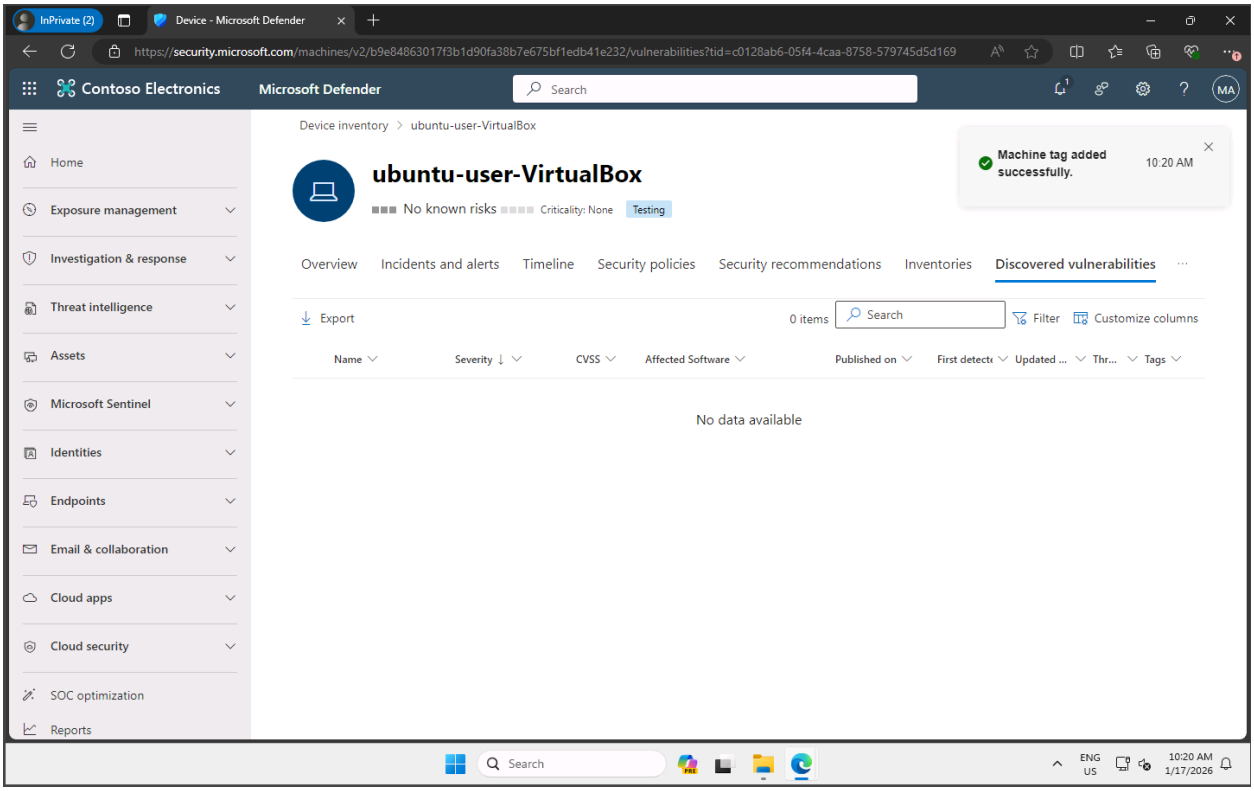


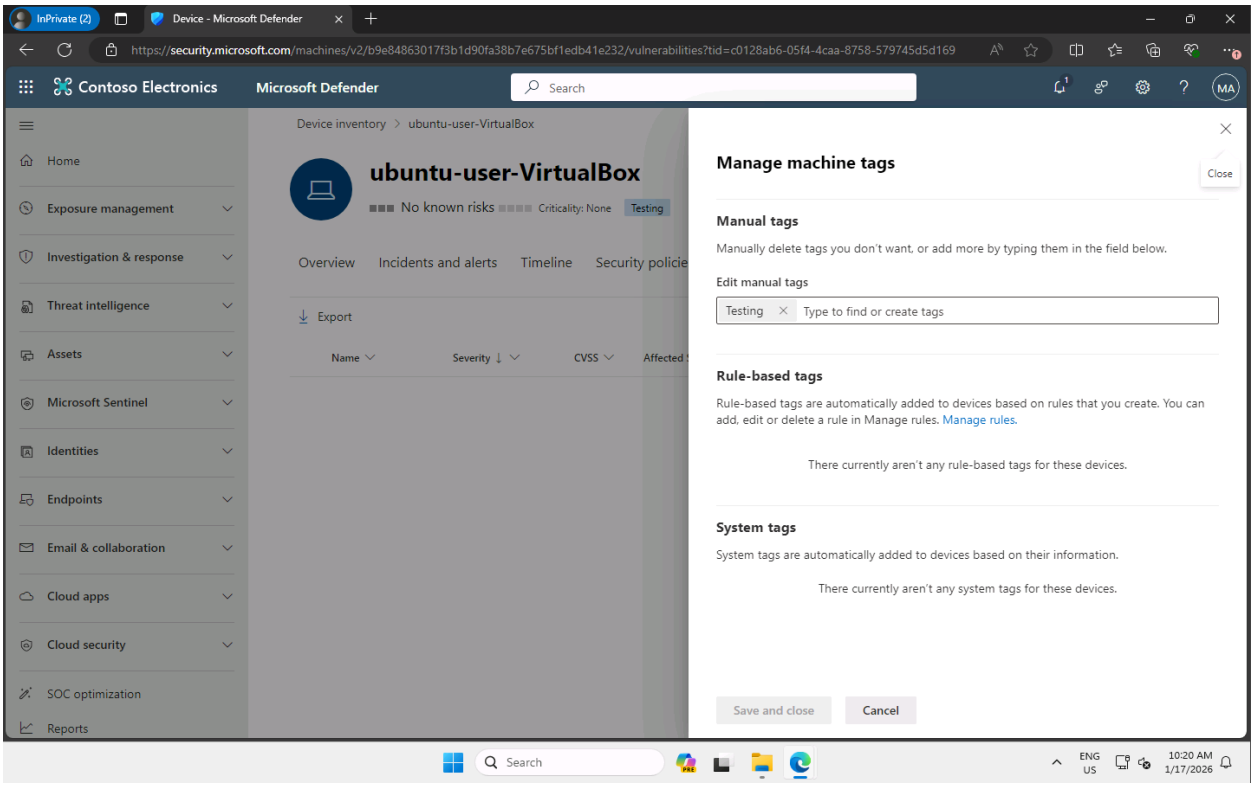
Validating in Defender portal

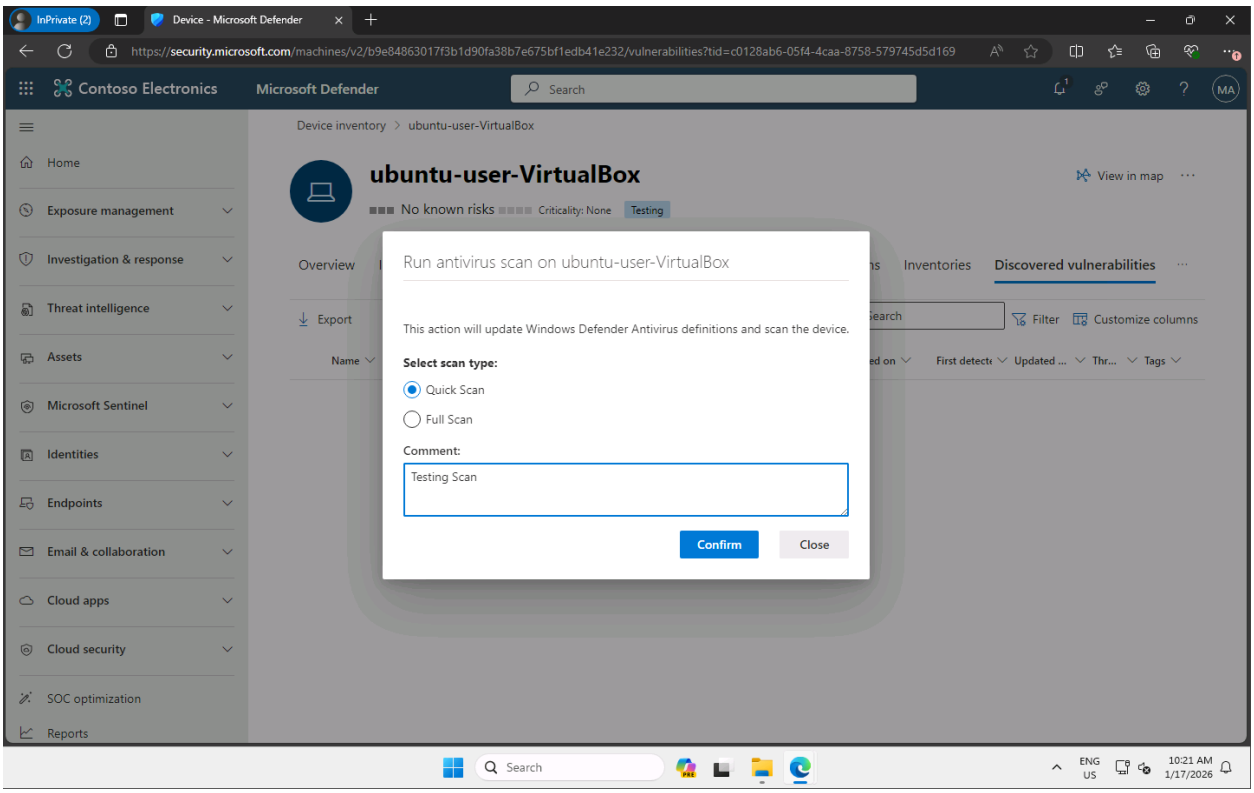

InPrivate (2)
Device - Microsoft Defender
https://security.microsoft.com/machines?filters=exclusionStates%3DIncluded%2CisTransient%3Dfalse&range=30&tid=c0128ab6-05f4-4caa-8758-579745d5...
Contoso Electronics
Microsoft Defender
Search
Home
Exposure management
Investigation & response
Threat intelligence
Assets
Devices
Identities
Applications
Cloud
AI agents
Microsoft Sentinel
Identities
Endpoints
Email & collaboration
Device Inventory
Create rules for devices
Classify critical assets
Assign criticality levels to your assets
Upgrade your vulnerability management capabilities
Try app control, baseline assessments, and more.
Transient devices have been automatically filtered out from some tabs to minimize noise. This filtering is determined by an internal algorithm, which mainly depends on the frequency of appearances of these discovered devices. To disable this automatic filtering, navigate to the filter menu.
All devices
Computers & Mobile
Network devices
IoT/OT devices
Uncategorized devices
Total 1
Critical assets 0
High risk 0
High exposure 0
Not onboarded 0
Newly discovered 1
Export
Search
30 Days
Customize columns
Filter
Filters: Transient device: No
Exclusion state: Not Excluded
Name
IP
Criticality level
Device category
Device type
Domain
Device AAD id
ubuntu-user-VirtualBox
10.0.2.15
Computers and Mo...
Workstation
Workgroup
ENG US
10:13 AM 1/17/2026


InPrivate (2)
Device - Microsoft Defender
https://security.microsoft.com/machines/v2/b9e84863017f3b1d90fa38b7e675bf1edb41e232/overview?tid=c0128ab6-05f4-4caa-8758-579745d5d169
Contoso Electronics
Microsoft Defender
Search
Device Inventory > ubuntu-user-VirtualBox
ubuntu-user-VirtualBox
No known risks
Criticality: None
View in map
Overview
Incidents and alerts
Timeline
Security policies
Security recommendations
Inventories
Discovered vulnerabilities
20-72-7F
10:08:31 AM
Last seen
Jan 17, 2026 10:10:28 AM
Onboarding status
Onboarded
Resources
-
Asset group
-
Device id
b9e84863017f3b1d90fa38b7e675bf1edb41e232
Defender engine version
101.25102.0005
Sense client version
Endpoint DLP status
Active alerts (Last 180 days)
No active alerts or incidents
Logged on users (Last 30 days)
Not supported
There were no logged on users during the given time range
Tracking user sign in activity on this operating system is not supported
Security assessments
Exposure level: Info
0 active security recommendations
ENG US
10:15 AM 1/17/2026


Device - Microsoft Defender
https://security.microsoft.com/machines/v2/b9e84863017f3b1d90fa38b7e675bf1edb41e232/timeline?range=%7B"timeRangeType"%3A"7"%2C"fromDate"...
Contoso Electronics
Microsoft Defender
Search
Device Inventory > ubuntu-user-VirtualBox
ubuntu-user-VirtualBox
No known risks
Criticality: None
View in map
Overview
Incidents and alerts
Timeline
Security policies
Security recommendations
Inventories
Discovered vulnerabilities
Aug 2025
Sep 2025
Oct 2025
Nov 2025
Dec 2025
Jan 2026
Maximize
Copy to clipboard
Export
Search
1 selected
1 Week
Customize columns
Filter
Event time
Event
Additional information
Load newer results
Jan 17, 2026 10:12:53.590 AM
File ingress from 2a00:1828:1000:2497::2 via curl: '/tmp/eicar.com.txt'
T1071: Application Layer Protoco
Jan 17, 2026 10:10:28.661 AM
A script was observed
Jan 17, 2026 10:10:28.660 AM
networkmanager failed to establish connection with 2620:2d:4000:1::2a:80
Jan 17, 2026 10:10:28.660 AM
networkmanager failed to establish connection with 2620:2d:4000:1::96:80
Jan 17, 2026 10:10:28.660 AM
networkmanager failed to establish connection with 2620:2d:4000:1::98:80
Jan 17, 2026 10:10:28.660 AM
networkmanager failed to establish connection with 2620:2d:4000:1::23:80
Home
Exposure management
Investigation & response
Threat intelligence
Assets
Devices
Identities
Applications
Cloud
AI agents
Microsoft Sentinel
Identities
Endpoints
Email & collaboration
ENG
US
10:16 AM
1/17/2026


Device - Microsoft Defender
https://security.microsoft.com/machines/v2/b9e84863017f3b1d90fa38b7e675bf1edb41e232/vulnerabilities?tid=c0128ab6-05f4-4caa-8758-579745d5d169
Contoso Electronics
Microsoft Defender
Search
Device inventory > ubuntu-user-VirtualBox
Machine tag added successfully.
10:20 AM
ubuntu-user-VirtualBox
No known risks
Criticality: None
Testing
Overview
Incidents and alerts
Timeline
Security policies
Security recommendations
Inventories
Discovered vulnerabilities
Export
0 items
Search
Filter
Customize columns
Name
Severity
CVSS
Affected Software
Published on
First detecte
Updated ...
Thr...
Tags
No data available
Home
Exposure management
Investigation & response
Threat intelligence
Assets
Microsoft Sentinel
Identities
Endpoints
Email & collaboration
Cloud apps
Cloud security
SOC optimization
Reports
ENG
US
10:20 AM
1/17/2026


InPrivate (2)
Device - Microsoft Defender
https://security.microsoft.com/machines/v2/b9e84863017f3b1d90fa38b7e675bf1edb41e232/vulnerabilities?tid=c0128ab6-05f4-4caa-8758-579745d5d169
Contoso Electronics
Microsoft Defender
Search
Home
Exposure management
Investigation & response
Threat intelligence
Assets
Microsoft Sentinel
Identities
Endpoints
Email & collaboration
Cloud apps
Cloud security
SOC optimization
Reports
Device inventory > ubuntu-user-VirtualBox
ubuntu-user-VirtualBox
No known risks
Criticality: None
Testing
Overview
Incidents and alerts
Timeline
Security policie
Export
Name
Severity
CVSS
Affected
Manage machine tags
Close
Manual tags
Manually delete tags you don't want, or add more by typing them in the field below.
Edit manual tags
Testing
Type to find or create tags
Rule-based tags
Rule-based tags are automatically added to devices based on rules that you create. You can add, edit or delete a rule in Manage rules. Manage rules.
There currently aren't any rule-based tags for these devices.
System tags
System tags are automatically added to devices based on their information.
There currently aren't any system tags for these devices.
Save and close
Cancel
ENG
US
10:20 AM
1/17/2026


InPrivate (2)
Device - Microsoft Defender
https://security.microsoft.com/machines/v2/b9e84863017f3b1d90fa38b7e675bf1edb41e232/vulnerabilities?tid=c0128ab6-05f4-4caa-8758-579745d5d169
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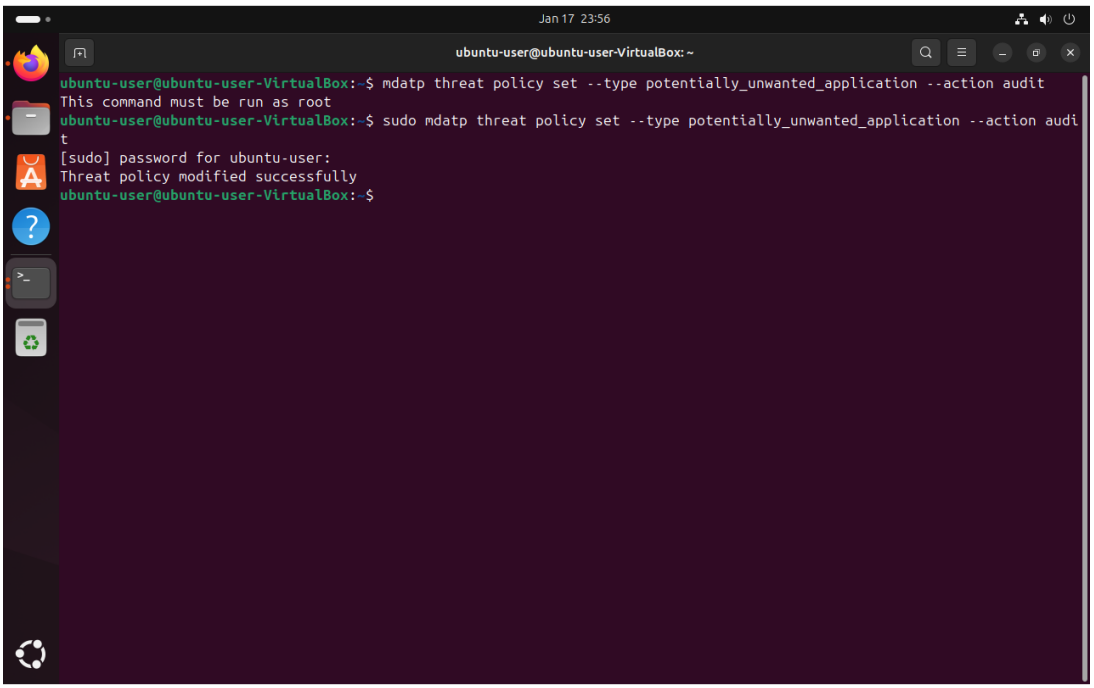
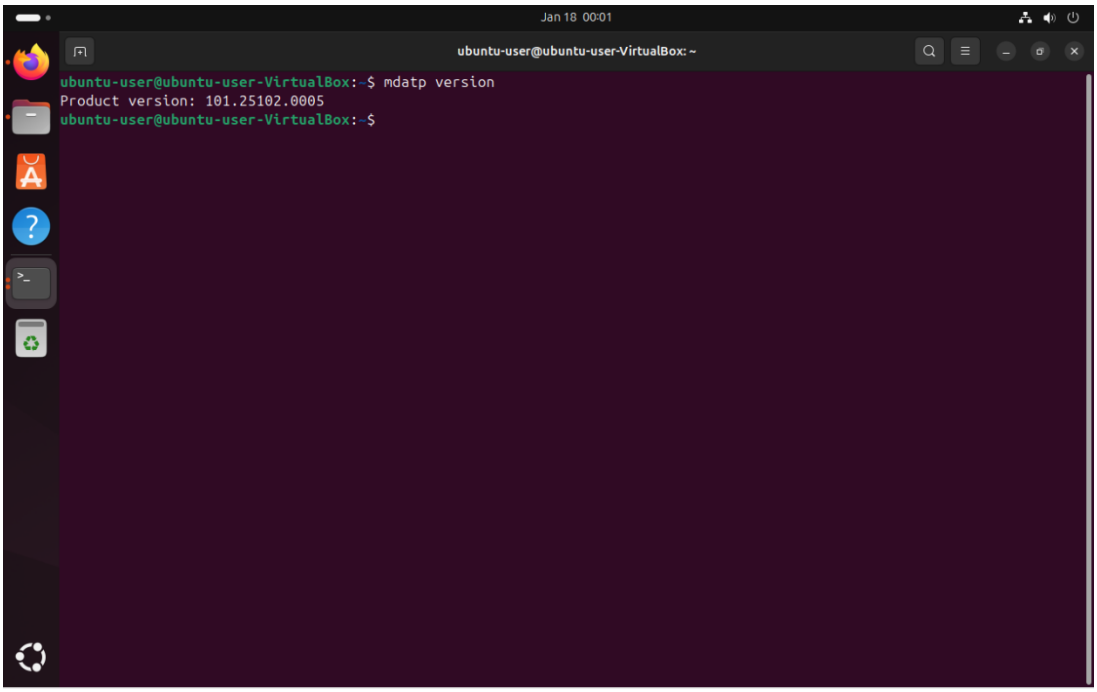
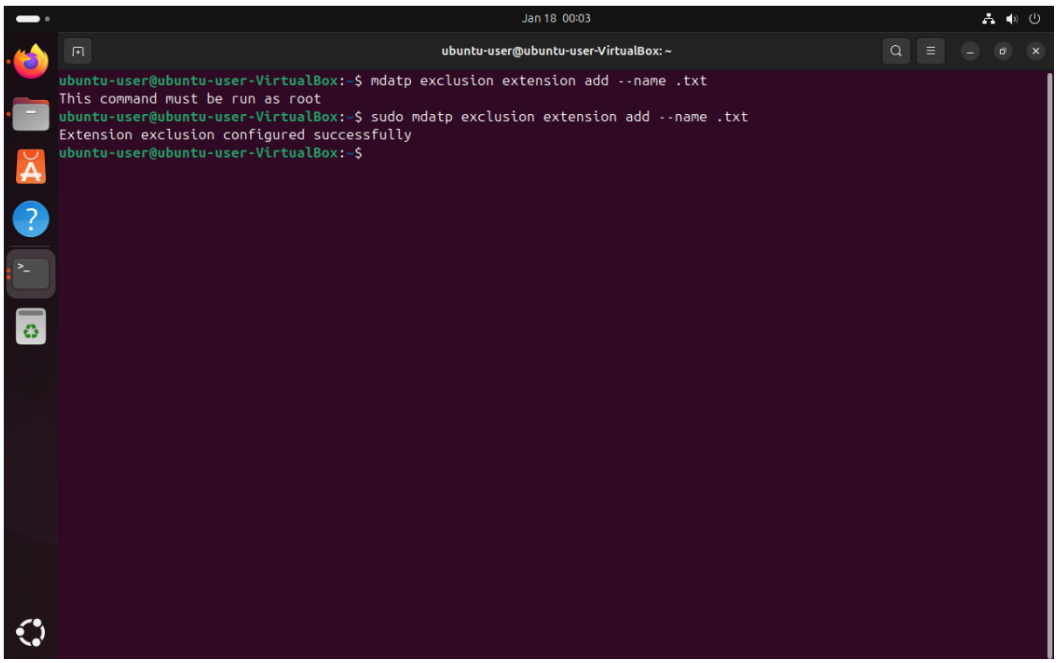
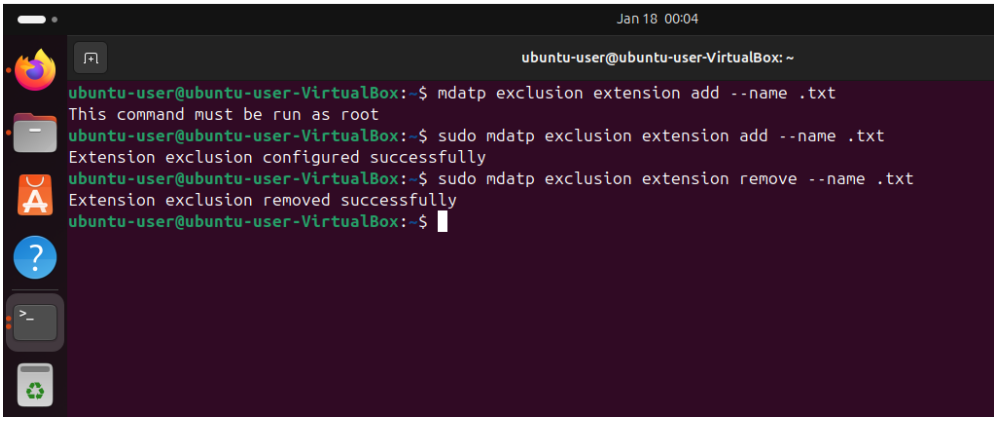
Reports
Device inventory > ubuntu-user-VirtualBox
ubuntu-user-VirtualBox
No known risks
Criticality: None
Testing
View in map
Overview
Inventories
Discovered vulnerabilities
Export
Filter
Customize columns
Name
First detecte
Updated ...
Thr...
Tags
Run antivirus scan on ubuntu-user-VirtualBox
This action will update Windows Defender Antivirus definitions and scan the device.
Select scan type:
Quick Scan
Full Scan
Comment:
Testing Scan
Confirm
Close
ENG
US
10:21 AM
1/17/2026

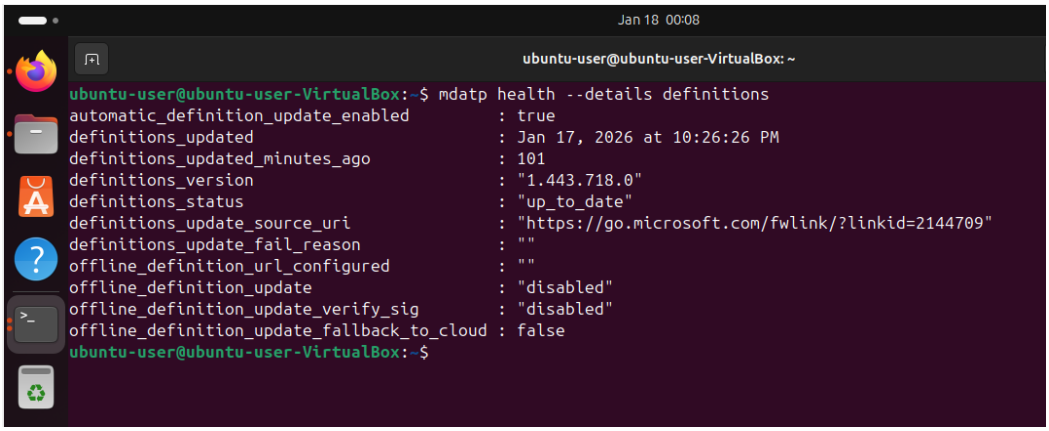

Jan 17 23:56
ubuntu-user@ubuntu-user-VirtualBox: ~
ubuntu-user@ubuntu-user-VirtualBox:~$ mdatp threat policy set --type potentially_unwanted_application --action audit
This command must be run as root
ubuntu-user@ubuntu-user-VirtualBox:~$ sudo mdatp threat policy set --type potentially_unwanted_application --action audit
[sudo] password for ubuntu-user:
Threat policy modified successfully
ubuntu-user@ubuntu-user-VirtualBox:~$

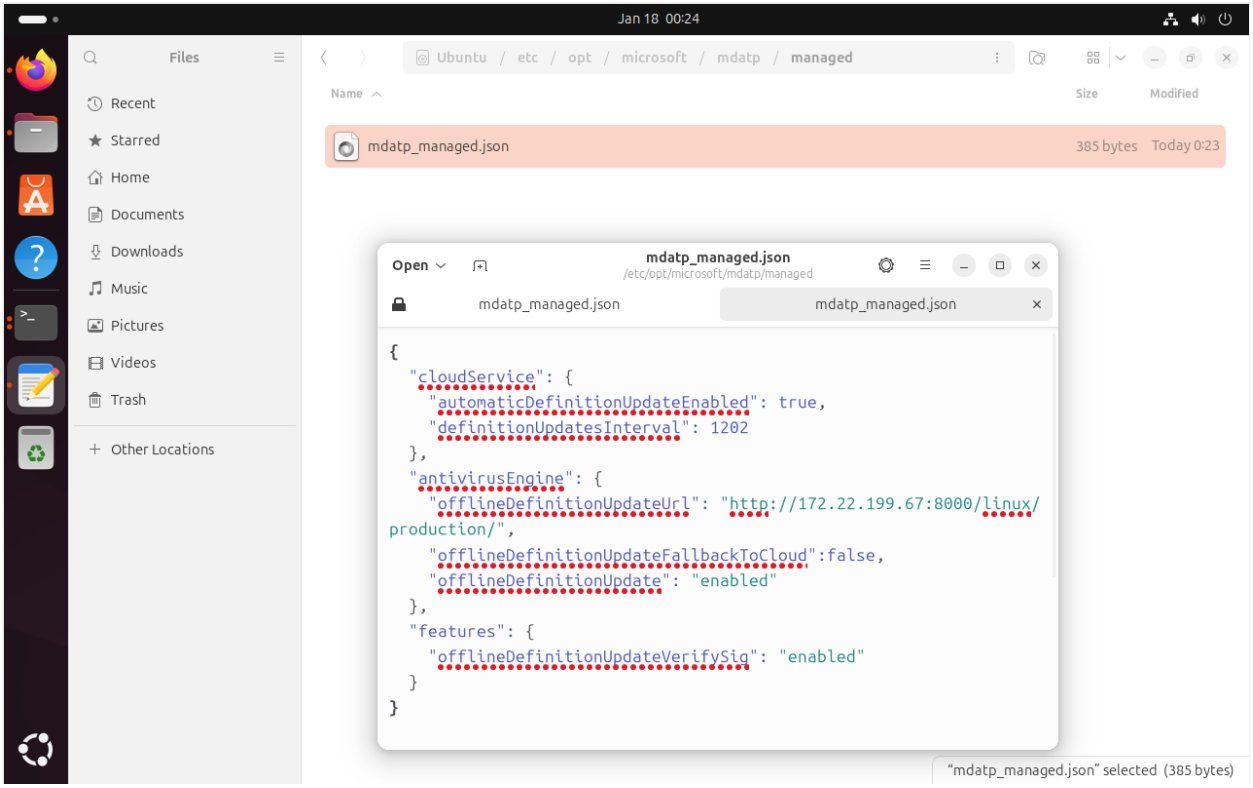

Jan 18 00:01
ubuntu-user@ubuntu-user-VirtualBox: ~
ubuntu-user@ubuntu-user-VirtualBox:~$ mdatp version
Product version: 101.25102.0005
ubuntu-user@ubuntu-user-VirtualBox:~$

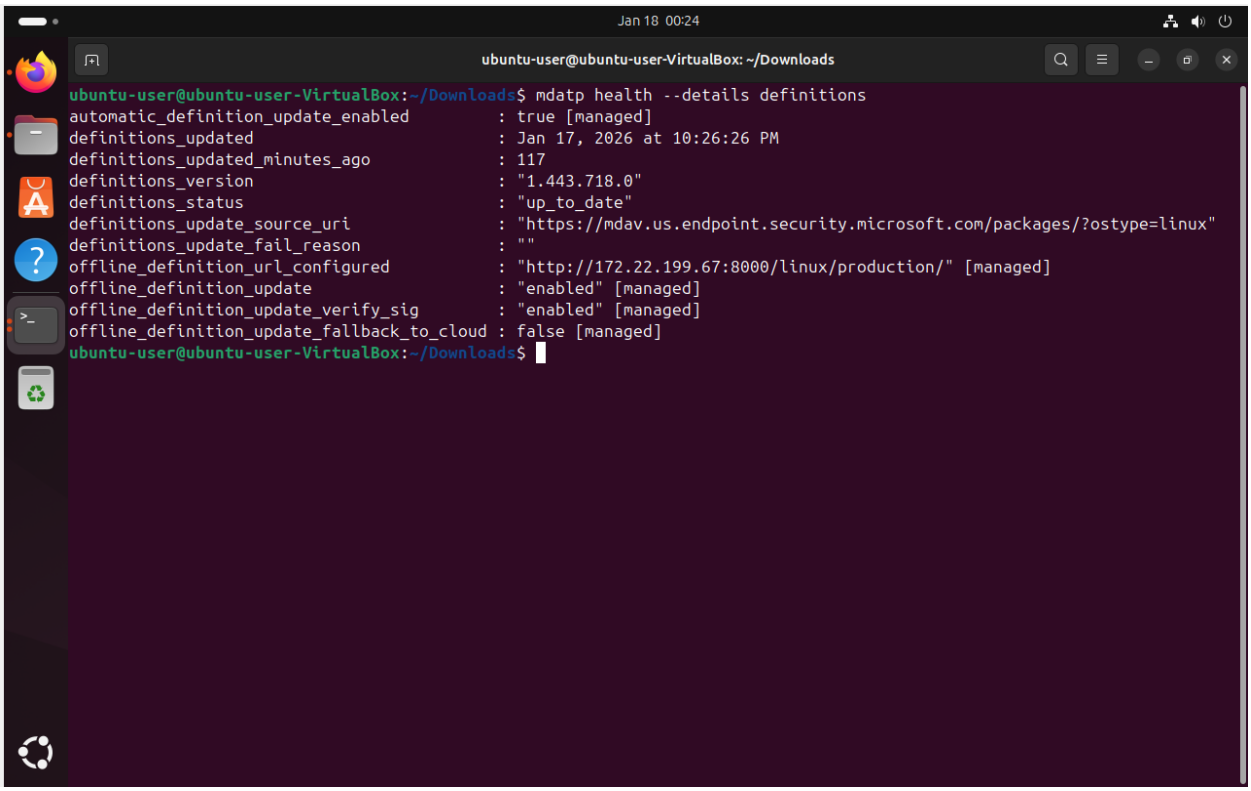


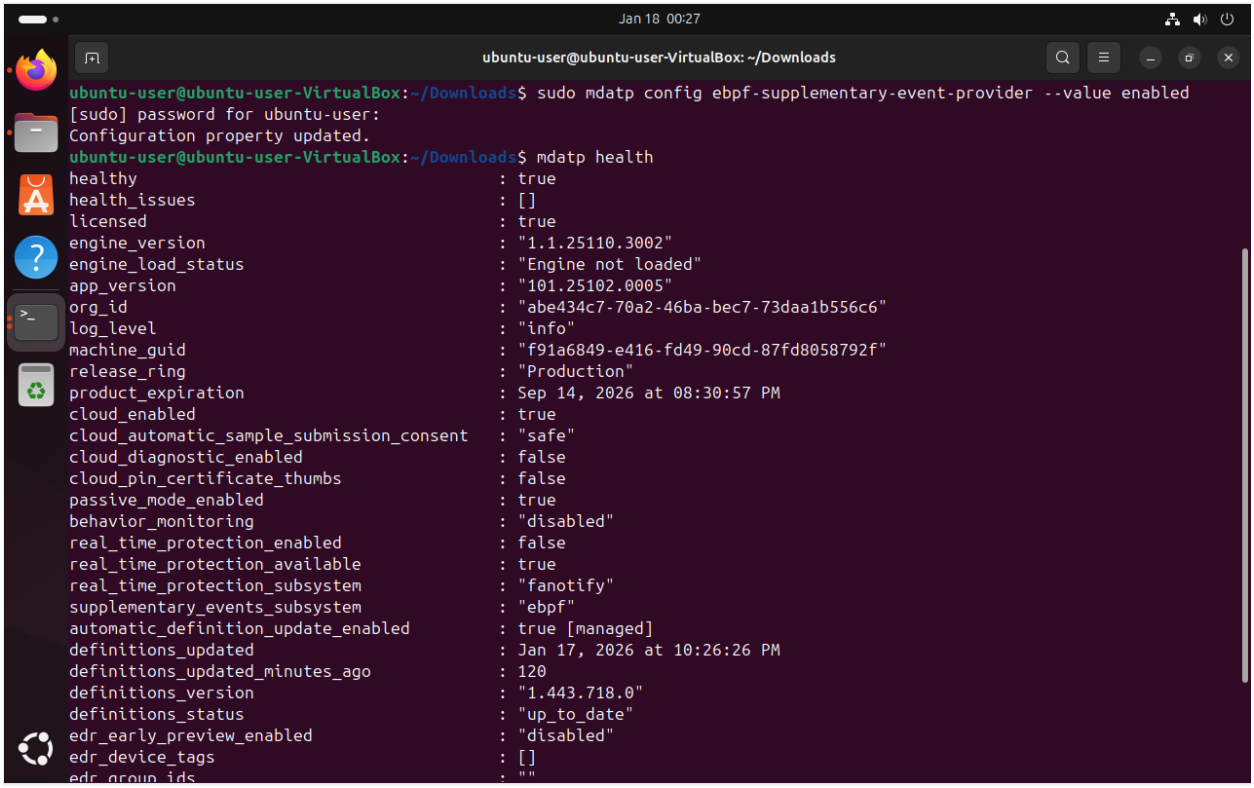
Adding exclusions






Jan 18 00:24
Files
Recent
Starred
Home
Documents
Downloads
Music
Pictures
Videos
Trash
Other Locations
Ubuntu / etc / opt / microsoft / mdatp / managed
Name
Size
Modified
mdatp_managed.json
385 bytes
Today 0:23
Open
mdatp_managed.json
/etc/opt/microsoft/mdatp/managed
mdatp_managed.json
mdatp_managed.json
{
"cloudService": {
"automaticDefinitionUpdateEnabled": true,
"definitionUpdatesInterval": 1202
},
"antivirusEngine": {
"offlineDefinitionUpdateUrl": "http://172.22.199.67:8000/linux/production/",
"offlineDefinitionUpdateFallbackToCloud":false,
"offlineDefinitionUpdate": "enabled"
},
"features": {
"offlineDefinitionUpdateVerifySig": "enabled"
}
}
"mdatp_managed.json" selected (385 bytes)


Jan 18 00:24
ubuntu-user@ubuntu-user-VirtualBox: ~/Downloads
ubuntu-user@ubuntu-user-VirtualBox:~/Downloads$ mdatp health --details definitions
automatic_definition_update_enabled : true [managed]
definitions_updated : Jan 17, 2026 at 10:26:26 PM
definitions_updated_minutes_ago : 117
definitions_version : "1.443.718.0"
definitions_status : "up_to_date"
definitions_update_source_uri : "https://mdav.us.endpoint.security.microsoft.com/packages/?ostype=linux"
definitions_update_fail_reason : ""
offline_definition_url_configured : "http://172.22.199.67:8000/linux/production/" [managed]
offline_definition_update : "enabled" [managed]
offline_definition_update_verify_sig : "enabled" [managed]
offline_definition_update_fallback_to_cloud : false [managed]
ubuntu-user@ubuntu-user-VirtualBox:~/Downloads$



10. References

- Microsoft Documentation: *Microsoft Defender for Endpoint on Linux*
- Deployment Guide: *Manual deployment for Microsoft Defender for Endpoint on Linux*

11. Abbreviations and Acronyms

Acronym	Full Form
MDE	Microsoft Defender for Endpoint
EDR	Endpoint Detection and Response
OS	Operating System